

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Requirements

ShopEZ — MERN Stack E-commerce Application

Phase: Requirement Analysis Phase

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Solution Requirements

1. Introduction

This document specifies the complete set of **Functional** and **Non-Functional Requirements** for the ShopEZ e-commerce platform. Each requirement is assigned a unique identifier to enable traceability from inception through design, implementation, and User Acceptance Testing (UAT).

2. Functional Requirements

FR-01 to FR-05 — User Management

Req. ID	Requirement Description	Priority	USN
FR-01	The system shall permit new users to register by providing a unique email address, full name, and password. Email must be validated for format correctness.	Must Have	USN-1
FR-02	The system shall authenticate registered users via email and password credentials and issue a signed JWT token upon success.	Must Have	USN-2
FR-03	The system shall store JWT tokens exclusively in HTTP-only cookies to prevent client-side JavaScript access (XSS mitigation).	Must Have	USN-2
FR-04	The system shall enforce Role-Based Access Control (RBAC) distinguishing between the <code>user</code> and <code>admin</code> roles.	Must Have	USN-0
FR-05	Authenticated users shall be able to update their profile including name, password, phone number, and saved shipping addresses.	Should Have	USN-1

FR-06 to FR-10 — Product Catalogue

Req. ID	Requirement Description	Priority	USN
FR-06	The system shall display a paginated catalogue of products presenting name, image, description, price, rating, and stock availability.	Must Have	USN-3
FR-07	The system shall provide full-text search allowing customers to query products by name and description keywords.	Must Have	USN-3
FR-08	The system shall support dynamic filtering by Category, Price Range, and Average Star Rating.	Must Have	USN-3
FR-09	The system shall support sorting by: Newest First, Price Ascending, Price Descending, and Highest Rated.	Should Have	USN-3
FR-10	Administrators shall be able to Create, Read, Update, and Delete product records, including uploading images via Cloudinary.	Must Have	USN-8

FR-11 to FR-15 — Shopping Cart & Checkout

Req. ID	Requirement Description	Priority	USN
FR-11	The system shall allow authenticated users to add products to a persistent shopping cart stored in MongoDB.	Must Have	USN-4
FR-12	The system shall allow users to modify item quantities and remove items; the cart total shall recalculate in real time.	Must Have	USN-4
FR-13	The system shall present a guided, multi-step checkout: (1) Shipping Address → (2) Payment Method → (3) Order Review & Placement.	Must Have	USN-5
FR-14	The system shall integrate with a Payment Gateway (Stripe / Razorpay) to securely process card and UPI-based transactions.	Must Have	USN-6
FR-15	Upon successful payment, the system shall create an order record, deduct purchased quantities from stock, and clear the user's cart.	Must Have	USN-6

3. Non-Functional Requirements

Req. ID	Category	Requirement Description	Priority
NFR-01	Security	All user passwords must be hashed using Bcrypt (salt rounds ≥ 10) prior to persistence.	Must Have
NFR-02	Security	All API routes that modify data must require a valid JWT; unauthenticated requests shall return HTTP 401.	Must Have
NFR-03	Security	Admin-only routes shall apply an additional <code>isAdmin</code> middleware check; non-admin access returns HTTP 403.	Must Have
NFR-04	Performance	Product catalogue API responses must be returned within 500ms under normal load conditions.	Should Have
NFR-05	Performance	All product images must be served via the Cloudinary CDN to ensure sub-second image load times globally.	Should Have
NFR-06	Usability	The frontend must be fully responsive from 320px (mobile) to 1920px (desktop).	Must Have
NFR-07	Reliability	MongoDB's document atomicity must ensure stock deduction and order creation are performed consistently, preventing overselling.	Must Have

Req. ID	Category	Requirement Description	Priority
NFR-08	Maintainability	All backend code must follow MVC pattern: Router → Middleware → Controller → Model. No business logic in route handlers.	Should Have
NFR-09	Scalability	The stateless JWT model shall ensure the API can be horizontally scaled without shared session state.	Could Have

4. Requirements Traceability Matrix

Req. ID	Design Artefact	Implementation File	UAT Test Case
FR-01, FR-02, FR-03	Solution Architecture	userController.js	UAT-001, UAT-002
FR-06, FR-07, FR-08	Data Flow Diagram	productController.js	UAT-005, UAT-006
FR-11, FR-12	ER Diagram	Cart.js model	UAT-008, UAT-009
FR-13, FR-14, FR-15	User Flow Sequence	orderController.js	UAT-011, UAT-012
FR-10	Solution Architecture	productController.js	UAT-013
NFR-01, NFR-02	Solution Architecture	authMiddleware.js	UAT-004, UAT-016